

TEAM KINYETI

Project Title: Optimizing RAG Document Retrieval for Agronomic Advice **Values-Led Problem Statement**

Smallholder farmers, agricultural extension workers, and rural advisory programs face a persistent barrier: reliable agronomic knowledge exists in manuals and research reports, but keyword-based search cannot surface it in response to real farmer questions. This gap slows decision-making on planting, pest control, and soil health, disproportionately affecting communities with limited access to trained agronomists. Team Kinyeti is building an optimized Retrieval-Augmented Generation (RAG) pipeline that uses semantic and hybrid retrieval to connect farmers' natural-language questions to the most relevant, trustworthy agricultural documents. Our solution is guided by accessibility, transparency, and accuracy—ensuring retrieved information is traceable to credible sources rather than opaque AI guesses—and by inclusivity, designing for users with varying technical literacy. Success means extension officers and farmers can find dependable answers faster, strengthening informed agricultural decisions and expanding equitable access to agronomic knowledge across underserved communities.

Word count: 139 words